Predictive Analysis of Patient Readmission Risk using Multivariate Statistical Models

Introduction :

Hospital readmission refers to a patient being admitted again within a short period (usually 30 days) after discharge. High readmission rate indicate :

- Poor Quality of Care
- Ineffective Treatment
- Increased healthcare cost

So, predictive which patients are at high risk of readmission help hospitals :

- Take preventive actions
- Improve patient Care
- Reduce unnecessary cost

Problem Statement :

The goal of this case study is :

To predict whether the patient will be readmitted or not using multiple patient - related factors.

This is a classification Problem :

Output = Yes (readmitted) / No (not readmitted)

What is Multivariate Statistical Modeling ?

Multivariate means using multiple Variable (features) at the Same time.

Example of Variables :

- Age
- Gender
- Blood pressure
- Previous admission

These Variables are analyzed together to find pattern

Data Collection .

Data is Collected from hospital records such as :

- Patient demographics
- Medical history (diseases, conditions)
- Lab Test result.

eg : Age, Blood pressure, Glucose level, Number of prior Visit.

Data preprocessing :

Before Modeling, data must be cleared.

Handling Missing Values :

- Fill with mean/ median or remove rows

Encoding Categorical Data :

Convert text (Male/Female) → numbers (0/1)

Feature Scaling

Normalize Values to bring them to Same rang

Removing Outliers

Remove abnormal Values.

Exploratory Data Analysis (EDA):

EDA helps to understand the relationship between Variables.

Techniques : ⊙ Correlation Matrix
⊙ Histograms
⊙ Box plots

eg :
Patients with high glucose → high readmission risk.

Multivariate Statistical Models Used :

1. Logistic Regression

Used when the Output is binary (Yes/No)

It predicts probability :

Output between 0 and 1

If the probability has > 0.5 → Readmitted

Key Idea :
Find relationship between multiple Variable and Outcome.

2. Decision Tree

- Split data based on Condition
- Easy to understand

If Age > 60 and Diabetes = Yes → High risk

3. Random Forest

Combination of multiple decision trees

More accurate and reduce overfitting

4. Support Vector Machine (SVM):

Separate class using a boundary (hyperplane)

Works well for complex datasets.

Model Training:

1. Split data into
   - Training Set (80%)
   - Testing Set (20%)
2. Train Model using training data
3. Test model on unseen data

Model Evaluation:

To check performance:

Accuracy How many predictions are correct

Precision Correct Positive Predictions

Recall } Ability of precision and recall

F1 Score }

Confusion matrix :

Shows :

- True Positive
- False Positive
- True Negative
- False Negative

Results and Interpretation :

Example Findings :

- Elderly patients have high readmission rate
- Patients with chronic diseases are more likely to return.
- Longer hospital stay increases probability

The Model helps identify high-risk patients early.

Applications :

- Hospital management System
- Patient Monitoring
- Insurance risk analysis
- Health Care policy Making

Advantages :

- Improves patient care
- Reduces hospital cost
- Helps doctors make better decisions
- Early risk detection

Limitations :

- Requires high - Quality data
- Missing or biased data affects accuracy
- Complex models are hard to interpret

Conclusion :

This case study shows how multivariate statistical models can be used to :

- Analyze multiple patient factors
- Predict readmission risk
- Support better healthcare decision

Predictive analysis of patient readmission using multivariate models plays a crucial role in improving healthcare Quality and reducing operational costs by enabling early identification of high - risk patients